MTS 102 CHEAT SHEET

Test-Ready Summary

Compiled by Felix the Astro-Coder

1. Limits

lim(x→a) f(x) = L → Plug in directly unless indeterminate

At infinity: Divide numerator and denominator by highest power of x

Special: lim(x→0) (sin x)/x = 1

2. First Principles (Definition of Derivative)

$f'(x) = \lim(h\to0) [(f(x + h) - f(x)) / h]$

3. Differentiation Rules

Power Rule: $d/dx(x^n) = nx^{n-1}$

Product Rule: $d/dx(uv) = u'v + uv'$

Quotient Rule: $d/dx(u/v) = (u'v - uv') / v^2$

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Chain Rule: $dy/dx = dy/du \times du/dx$

4. Maxima, Minima & Points of Inflection

Set $f'(x) = 0$ → Turning points

Use second derivative $f''(x)$:

- If $f''(x) > 0$ → Minimum
- If $f''(x) < 0$ → Maximum
- If $f''(x) = 0$ → Possible inflection (check signs before & after)

5. Integration Rules

$\int x^n \, dx = x^{n+1} / (n + 1) + C$ (for $n \neq -1$)

$\int (1/x) \, dx = \ln|x| + C$

$\int e^x \, dx = e^x + C$

$\int a \cdot f(x) \, dx = a \cdot \int f(x) \, dx$

6. Definite Integration

$\int_a^b f(x) \, dx = F(b) - F(a)$

7. Volume of Revolution (about x-axis)

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$V = \pi \int_a^b [f(x)]^2 \, dx$

8. Curve Sketching Tips

Find intercepts (x and y)

Solve $f'(x) = 0$ → Turning points

Use $f''(x)$ to determine the nature of points

Check for symmetry (even/odd)

Sketch smooth curve using key feature

9. Quick Integrals Reference

Function	Integral
x^n	$x^{n+1} / (n + 1) + C$
e^x	$e^x + C$
$1/x$	$\ln$
$\cos(x)$	$\sin(x) + C$
$\sin(x)$	$-\cos(x) + C$

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Wishing you success in your test!

You've got this

Astronaut Felix

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